

James Beavan

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SUMMARY

Junior at Georgia Tech (Mathematics, Data Science concentration) focused on data engineering. Skilled in architecting data pipelines, designing database schemas, and building backend APIs using Python, SQL, Dagster, and cloud infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Go

Data Engineering: Dagster, DuckDB, PostgreSQL, Databricks, Pandas, NumPy

Cloud & Infrastructure: AWS (EC2, S3, Route 53), Oracle Cloud (OCI), Docker, Linux/Bash, Git

ML & Analytics: XGBoost, scikit-learn, LangChain, Tableau

APIs & Apps: FastAPI, RESTful APIs, Dash, React

WORK EXPERIENCE

Data Insights Intern

Jun. 2026 - Aug. 2026

Northrop Grumman

Falls Church, VA

- Developed Python scripts to automate documentation generation for 50+ SQL objects (stored procedures, views) across ~20 manufacturing sites
- Diagnosed and resolved conflicting aggregation logic across two dashboard views, aligning metric definitions and eliminating reporting discrepancies
- Optimized a core Tableau dashboard, cutting filter time by 25%
- Won CIDO Intern Hackathon with an app to track and filter international procurement data for \$3M+ orders
- Built a two-stage documentation chatbot using LangChain and OpenAI with dynamic file routing and context filtering, optimizing prompts and token allocation to reduce irrelevant context

Software & Data Intern

May 2025 - May 2026

Lindaben Foundation

Columbia, MD

- Built and deployed a Dagster + DuckDB + Docker pipeline on AWS EC2 processing 180K records, with incremental loads, timestamp-based deduplication, and seven aggregation tables powering an analytics dashboard
- Built and documented 30+ REST API endpoints with filtering, pagination, and sorting, plus role-based access control across five user roles
- Developed a GPT-4-assisted migration tool converting legacy Excel delivery files into relational JSON
- Analyzed Excel time logs with Python, Pandas, and Openpyxl, detecting 300+ data entry errors and visualizing them as heat maps

Data Science Intern

Sep. 2024 - Jan. 2025

Intechgambit

Falls Church, VA

- Stress-tested a Python/Biopython-based tool for retrieving gene sequences from NCBI, identifying its upper limit at 40,000 entries before failure
- Evaluated AI models (ChatGPT, Gemini, Copilot) by testing 50+ prompts; recorded accuracy and runtime metrics to compare performance across tasks

PROJECTS

Stockidence | stockidence.com | [GitHub](https://github.com)

Aug. 2026 - Present

- Built an on-demand stock research platform on Dagster and DuckDB, incrementally ingesting prices, fundamentals, and news sentiment from 4 APIs behind per-endpoint staleness TTLs
- Designed a three-layer DuckDB warehouse (raw → staging → mart) that derives 11 technical indicators in-pipeline instead of calling premium API endpoints
- Trained an XGBoost LambdaMART (rank:ndcg) model on 15.9K quarterly rows (377 tickers); walk-forward validated over 26 out-of-sample quarters (rank IC +0.163; top 20 averaged +4.6 pp/qtr over the S&P 500, beating it in 62% of quarters)
- Served via FastAPI + React with Docker Compose on Oracle Cloud; automated quarterly retrain refuses to publish on stale data

EDUCATION

Georgia Institute of Technology | Atlanta, GA

Jul. 2025 - May 2028

B.S. in Mathematics, Data Science Concentration

GPA: 3.6/4.0

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Mathematics of Data Science, Statistics and Probability, Discrete Mathematics